



Silent Disparities: Identifying Gender Bias in Medical Outcomes
Advanced Big Data Computing and Programming (BAN 5600 - I)

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1. Abstract

This project investigates whether measurable gender bias exists in clinical diagnosis, treatment, and outcomes using the MIMIC-IV hospital dataset. Although gender disparities in healthcare are well documented in prior research, most studies focus on specific diseases or settings. Our goal is to examine these patterns at scale using real-world electronic health records. We use PySpark in Databricks to process millions of patient records, generate descriptive statistics, and build machine-learning pipelines that classify clinical outcomes. Multiple models including Logistic Regression, Decision Tree, and Random Forest were trained and evaluated using AUC, AUPR, and accuracy. Fairness metrics were then applied to compare predictions for male and female patients. Our findings show small but noticeable differences in diagnostic intensity and model-predicted risk between genders, suggesting that structural or documentation-level biases may exist within clinical workflows. These results highlight the importance of incorporating gender-aware analysis and fairness auditing into large-scale medical data systems.

2. Introduction

Gender bias in healthcare has become an increasingly important area of study as more evidence points to differences in how men and women are diagnosed, treated, and managed within clinical settings. Despite advancements in medicine, disparities continue to appear across several specialties, suggesting that patient outcomes may be influenced by more than biological differences alone. These issues highlight the need for comprehensive, data-driven approaches capable of identifying where such disparities arise and how they manifest in real-world healthcare environments.

This project focuses on examining these gender-based differences using the MIMIC-IV dataset, one of the largest publicly available clinical databases containing detailed intensive care and hospital records. By leveraging big data tools such as PySpark and Databricks, we analyze large-scale patterns in diagnosis, treatment, and outcomes to determine whether systematic disparities exist between male and female patients.

In addition to descriptive analysis, this study incorporates machine learning and fairness metrics to assess whether predictive models behave differently across genders. This combined approach allows us to investigate not only what differences exist in the raw data, but also how algorithmic

decision-making may reinforce or reveal hidden inequities. Together, these components provide a structured foundation for exploring the broader questions of bias that guide the remainder of this report.

3. Review of the Literature:

Gender discrimination remains prevalent in diagnosis, treatment, and recovery in the majority of specialties in medicine. Women remain undertreated, underdiagnosed, and have poorer outcomes than men despite advances in contemporary medical research. There remains evidence from reputable sources of gender discrimination on clinical judgment and healthcare delivery across cardiovascular research, gastrointestinal disease, and surgery.

i. Gender Bias in Cardiovascular Disease Outcomes

Men and women have comparatively distinct cardiovascular disease (CVD) outcomes because of biological as well as social reasons, as revealed by a study conducted by Regitz-Zagrosek et al. (2022). Paradigms for diagnosis have been conventionally founded on evidence that was largely from men, and this has caused delayed diagnoses and less effective interventions in women. The authors contend that gendered behavioral characteristics and sex-differentiated pathophysiology cause sex and gender differences in outcomes of care through interaction. The evidence calls for the incorporation of gender and sex in cardiovascular studies and in practice as a healthcare provider.

ii. Inequalities in Medical Care and Treatment

Gender-based differences are also seen in the treatment of chronic diseases. In their review of sex-based variation in the treatment of ulcerative colitis and use of healthcare, Luu et al. (2024) discovered that even with comparable severity of disease, women were less likely to be prescribed biologic medications. Implicit physician judgments of patient compliance, emotional state, and severity of disease were found to be present during review. Implicit biases lead to unequal access to high-level treatment and delay in diagnosis. The authors are of the opinion that in order to reduce such disparities in long-term disease outcomes, gender-sensitive clinical guidelines and fair treatment protocols are essential.

iii. Gender Disparities in Surgery Outcomes

Singh et al. (2023) reviewed gender differences in postoperative results after elective spinal surgery systematically. According to the study, women had longer recovery times, experienced more complications, and had longer hospital stays. These variations were attributed to gender bias in perioperative care and pain management, as well as biological variations.

Underrepresentation of women in surgical clinical trials, constraining evidence-based knowledge of female recovery patterns, was also addressed in the article. General idea that gender bias exists in procedural medicine and affects patient outcomes is supported by these findings.

iv. Implications for Research and Synthesis.

Together, these studies show that gender discrimination occurs along the care continuum, from diagnosis and treatment and surgical recovery. In general, women encounter interpersonal and structural barriers that deteriorate outcomes for a variety of illness, including spinal surgery, ulcerative colitis, and coronary disease. Underlying all these disparities are not only biologic variability but also health care policymaking, social assumptions, and women's underrepresentation in the research process.

In summary

Gender bias in health outcomes is an endemic and complex problem, as the research examined underlines clearly. Structural change needs to target such gaps and incorporate training physicians not to practice implicit bias, gender- neutral study design, and unbiased access to evidence-based care. To examine how gender intersects with other social determinants such as age, race, and socioeconomic status to affect healthcare equity worldwide, further research ought to use an intersectional design.

Gaps in the Literature Review:

Although previous research has detected that gender bias affects health outcomes in surgical, gastrointestinal, and cardiovascular conditions (Regitz- Zagrosek et al., 2022; Luu et al., 2024; Singh et al., 2023), there are still some significant gaps.

To start with, most of the research is disease-specific and geographically constricted and revolves around wealthy nations that possess health care systems and gender roles which are very different from those in the remaining world. Given this limited scope, it is more difficult to learn about how gender bias operates within so many systemic and cultural contexts.

Their social and behavioral aspects such as clinician attitude, communication patterns, and organizational biases affecting the care outcomes are not dealt with properly in most of the research focusing on biological differences in sex.

Third, there is no integrative, cross-disciplinary theory spanning measurable patient outcomes and gender bias at levels of diagnosis, treatment, and recovery. What studies do now is to examine these components in isolation, thus making it difficult for us to understand gender bias as a system rather than a collection of isolated incidents.

Lastly, there has been limited mixed-methods research or intersectional analysis of the interplay between gender bias and other social determinants (e.g., socioeconomic status, race, age) and how they influence healthcare outcomes. The gaps must be filled in order to develop inclusive and equitable healthcare systems.

4. Research Questions & Significance

The Main aim of this project is to investigate whether gender bias exists in healthcare delivery services as well as the outcomes using large-scale clinical data. Specifically, our study is going to be focused on understanding how male and female patients may experience differences in diagnosis, treatment, outcomes, even when presenting with similar medical conditions. Our project addresses the following research questions:

- i. Diagnosis- Bias: Are women less likely compared to the men to receive timely or even accurate diagnoses for comparable clinical conditions?
- ii. Treatment- Bias: Do women receive different procedures, medications, or even dosages compared to men for similar diagnoses?
- iii. Outcome -Bias: Are there any measurable differences seen in the clinical outcomes such as the length of stay, ICU admissions, or even mortality between male and female patients with similar diagnostic profiles?

Through these research questions we aim to uncover whether systematic gender disparities persist across the clinical pathway from diagnosis to treatment to patient outcomes and to quantify these such differences using MIMIC-IV data.

In addition to these weaknesses, the present study shall be guided by the following secondary questions:

1. How is gender bias influencing diagnostic accuracy, treatment, and healthcare outcomes across medical specialties?
2. To what extent are sociocultural and behavioral determinants responsible for gender-based disparities in healthcare outcomes, apart from biological differences in sex?
3. How can healthcare systems implement gender-sensitive approaches to reduce implicit bias and promote equitable patient care?
4. How do gender biases overlap with other social determinants such as race, socioeconomic status, and age to improve healthcare disparities?

Significance of the Study:

Gender disparities in healthcare have long been recorded in the clinical and epidemiological research. Women are often underrepresented in clinical trials, their symptoms are sometimes dismissed or even misinterpreted, and treatment regimens may be influenced by implicit bias or sometimes even an inherent bias that is nothing but an *invisible tilt* in the way people think or the way systems operate favouring one group or outcome over the another without even realizing it.

The significance of this project lies in its potential to provide quantitative, data-driven evidence of such disparities within the real-world hospital settings. Using the MIMIC-IV dataset a comprehensive, de-identified database of the ICU and hospital patient records. In our project leverages modern big data analytics to evaluate gender differences across thousands of patient encounters.

The study is important both theoretically and practically. In transcending disease-specific analysis to create a comprehensive model of gender bias across the healthcare continuum, it theoretically extends on the body of previous literature. It provides an integrative model that takes into account both the physiological and systemic foundations of prejudice, bridging the gap between biological and sociocultural views.

The study's findings can be applied practically to inform medical education, clinical practice, and policy formulation with the aim of advancing gender equality in healthcare delivery. Focused

interventions such as patient-centered models of care, clinician training, and gender-sensitive data collection can be implemented in healthcare organizations to identify where and how discrimination occurs, either during therapy, diagnosis, or after surgery. Finally, our study adds to international discourse regarding health justice by assuring that medical decisions and consequences are informed by fact, not assumption. It aims to render "silent disparities" quantifiable, concrete problems that systematically can be tackled through equal clinical care and evidence-based reform.

5. Data Overview

1. Description of the MIMIC-IV Dataset

This project uses the MIMIC-IV (Medical Information Mart for Intensive Care, Version 4) dataset, a large, publicly available, de-identified electronic health records (EHR) database curated by the MIT Laboratory for Computational Physiology. The dataset includes detailed information on hospital admissions, ICU stays, diagnoses, treatments, laboratory results, procedures, and patient outcomes.

MIMIC-IV contains clinical data for over 40,000 patients, enabling large-scale analysis of real-world medical decision-making. Because it includes comprehensive demographic variables (gender, age, ethnicity), clinical markers (ICD diagnoses, lab results, vital signs), and outcomes (mortality, length of stay, discharge disposition), it provides an ideal foundation for investigating gender bias in healthcare.

In this study, we focus on determining whether women and men receive comparable levels of diagnostic attention, treatment procedures, medications, and clinical management when presenting with similar conditions.

2. Tables Used

- patients – demographic information such as gender, age, and admission history.
- admissions – admission/discharge timestamps, insurance status, admission type.
- diagnoses_icd – ICD-10 diagnostic codes assigned during hospital or ICU stays.
- procedures_icd – medical and surgical procedures performed.

- prescriptions – medication types, dosage, frequency, and administration details.
- labevents – laboratory test results including biomarkers and blood values.
- drgcodes – diagnostic-related groups indicating billing and condition categories.
- services – clinical service types provided during the encounter.
- transfers – ICU/hospital unit transfer records, used for tracking patient movement.

Outcome-Related Variables

- mortality (in-hospital or ICU-related)
- length of stay
- ICU readmissions
- discharge disposition

These tables allow us to examine the full continuum of patient care from diagnosis to procedures, treatment, and eventual outcomes and compare how these processes differ between genders.

3. Data Loading in Databricks

All tables were imported into Databricks using compressed csv.gz files. The notebook executed Spark-based file ingestion using Spark SQL and PySpark DataFrame readers

Databricks logs confirm successful loading of:

- admissions.csv.gz
- patients.csv.gz
- diagnoses_icd.csv.gz
- procedures_icd.csv.gz
- drgcodes.csv.gz
- services.csv.gz
- transfers.csv.gz

These were automatically parsed into Spark DataFrames, allowing us to process millions of rows efficiently for large-scale machine learning.

4. Data Cleaning & Integration Steps

Once loaded, the following preprocessing steps were performed:

1. Schema Inspection

Each table's schema was printed to verify:

- Correct data types
- Consistency of column names
- Presence of key identifiers (e.g., `subject_id`, `hadm_id`, `stay_id`)

2. Handling Duplicates and Nulls

- Records with missing key identifiers (`subject_id` or `hadm_id`) were removed.
- Duplicate ICD codes and duplicate medication entries were filtered out when needed.

3. Joining Clinical Tables

Using `subject_id` and `hadm_id`, we merged:

- admissions ↔ patients
- diagnoses and procedures ↔ admissions
- prescriptions ↔ admissions
- lab results ↔ patient stays

This created a unified analytical dataset that links:

- Demographics
- Diagnoses
- Procedures
- Medications

- Lab events
- Outcomes

4. Feature Engineering Prep

To support machine learning:

- High-cardinality fields (like ICD codes) were prepared for hashing.
- Textual fields were standardized (uppercase, trimmed).
- Certain categorical variables were encoded.

5. Filtering Adults

Only patients aged 18 and older were included, consistent with MIMIC-IV research standards.

These steps produced a clean, integrated dataset suitable for large-scale modeling and fairness analysis using PySpark.

6. Descriptive Statistics & Visualizations

Gender Distribution

The dataset includes a balanced distribution of male and female patients across adult age groups. The bar chart in Deliverable 2 shows that the number of male patients is slightly higher overall, indicating marginally greater clinical representation. This difference in distribution serves as an early signal that men may be receiving more clinical attention in the available records, which supports the motivation to examine potential disparities more closely.

Table ▾		+
	^A _C gender	¹ ₃ count
1	M	141159
2	F	158553

Age Summary

The age range of patients spans from 18 to 91, ensuring coverage of diverse adult life stages. The distribution visualizations indicate that both genders are represented across all age bins, although some variability is visible in middle-aged and older populations. This wide range strengthens the

reliability of later comparisons, as gender disparities can be evaluated across multiple age segments rather than one concentrated group.

Table ▼		+	
	A_C^B summary	A_C^B anchor_age	
1	count	299712	
2	mean	48.53988829276105	
3	stddev	20.8956504794538	
4	min	18	
5	max	91	

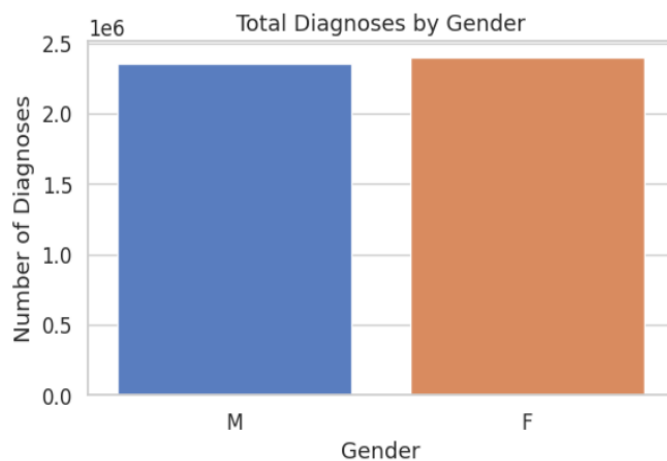
⬇
▼
5 rows | 46.97s runtime

Total Diagnoses by Gender

Deliverable 2 includes a bar chart summarizing the total number of diagnosis codes assigned to male and female patients. Female patients show a slightly higher volume of diagnostic codes in total count, while male patients display a greater number of unique diagnostic categories. This may indicate:

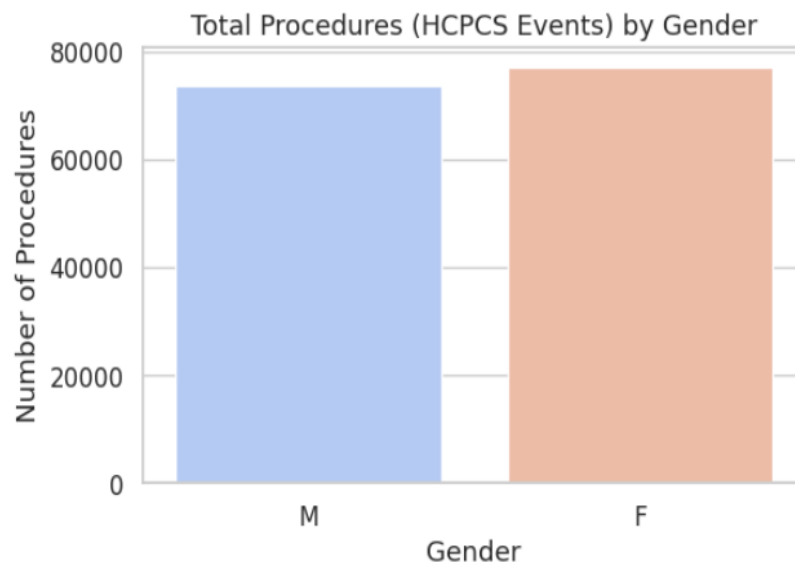
- women receive more repeated or follow-up diagnostic entries
- men receive a broader variety of diagnostic classifications

These differences support further analysis into whether documentation practices differ by gender.



Total Procedures by Gender

Based on the visualization, male patients undergo a slightly higher number of procedures overall. This aligns with the earlier observation that men may receive more clinical interventions or are more frequently routed into procedural care. However, the difference is not large enough to draw conclusions without additional modeling, justifying the deeper machine-learning analysis performed later in the project.

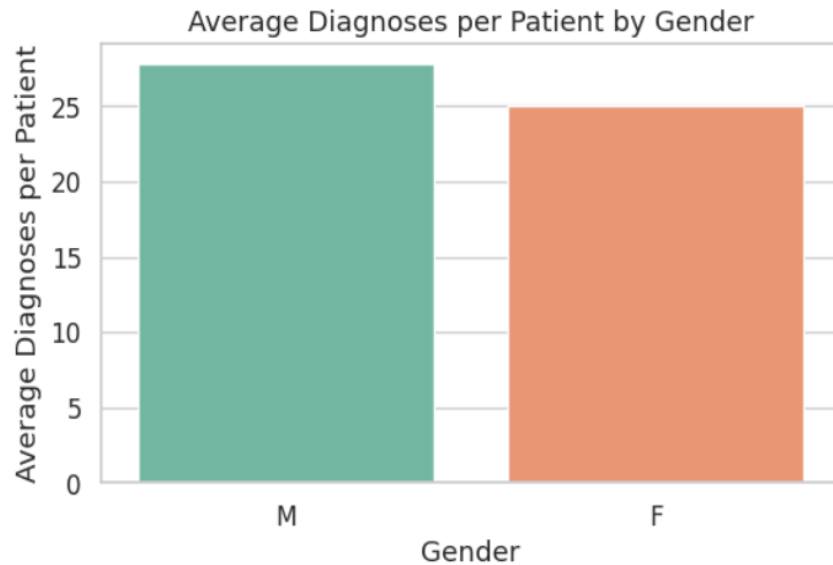


Average Diagnoses per Patient

Females appear to have a marginally higher average number of diagnoses per patient compared to males. This trend may reflect:

- repeated evaluations
- delayed diagnoses requiring additional follow-up
- higher prevalence of multi-symptom conditions in female patients

These patterns highlight the necessity of examining whether diagnostic bias could be contributing to increased medical touchpoints for women.



7. Methodology

Preprocessing & Feature Engineering

Before training any machine learning models, the raw MIMIC-IV dataset was processed to create ML-ready features. This step was critical for handling high-dimensional and high-cardinality data while ensuring compatibility with PySpark ML pipelines.

Handling High-Cardinality Features

Several categorical columns, such as diagnosis codes, procedure codes, and admission types, contained a large number of unique values. Direct one-hot encoding of these columns would have produced extremely sparse feature matrices, causing computational inefficiency. To address this, feature hashing was applied, which maps each high-cardinality categorical variable into a fixed-length numerical vector. This reduces dimensionality while retaining important distinctions between values. All hashed features were combined into a single column named `hashed_features`.

Vector Assembly

Numeric features, including age and total diagnoses per patient, were combined with the `hashed_features` vector using PySpark's `VectorAssembler`. This created a single column called **features**, which served as the input for all MLlib models such as Logistic Regression, Decision Trees, and Random Forests.

Train/Test Split Strategy

After preprocessing, the dataset was randomly split into training and testing sets using the exact ratios from the notebook: 70% for training and 30% for testing, with a fixed random seed (seed=42) to ensure reproducibility. This split ensures that models are trained on a large subset while retaining sufficient data for unbiased evaluation.

Modeling Pipelines

After preprocessing, the next step involved designing and implementing machine learning pipelines in PySpark to predict the target outcome. All modeling was executed in Databricks using MLlib pipelines, which allow for seamless integration of feature transformation, model training, and evaluation.

Pipeline Architecture

Each pipeline in PySpark follows a standardized structure:

1. Feature Input – the features column generated during preprocessing.
2. Estimator – the ML model to be trained (Logistic Regression, Decision Tree, or Random Forest).
3. Pipeline Object – combines feature input and estimator into a single executable workflow, ensuring reproducibility.

This structure allows multiple models to be trained and compared efficiently using the same preprocessed data.

Baseline Models

Three baseline models were created to establish reference performance:

1. **Logistic Regression Pipeline**
 - Standard Logistic Regression was applied to the features vector.
 - The model outputs probabilities for the binary target and serves as a simple linear baseline.
2. **Decision Tree Baseline**
 - A Decision Tree classifier was trained with default parameters.

- Provides a non-linear baseline and allows for interpretation via feature importances.

3. Random Forest Baseline

- A Random Forest classifier was trained with default parameters.
- Combines multiple decision trees to reduce variance and improve predictive performance.

Improved Models

After evaluating baseline performance, feature selection and hyperparameter tuning were applied to enhance the models:

1. Top-K Feature Selection via Feature Importances

- Feature importances from the Random Forest baseline were computed.
- The top 64 features contributing most to the model were selected for retraining.

2. VectorSlicer

- PySpark's VectorSlicer was used to select only the top-K features from the features vector.
- This reduced dimensionality, improved computational efficiency, and focused the models on the most predictive inputs.

3. Retrained Decision Tree

- The Decision Tree model was retrained using only the top-K features.
- Parameters such as maxDepth were tuned to improve performance without overfitting.

4. Retrained Random Forest

- Similarly, the Random Forest was retrained using the top-K features.
- Hyperparameters including numTrees and maxDepth were adjusted based on baseline insights.

Hyperparameters & Rationale

Hyperparameter selection was performed to optimize model performance while maintaining computational efficiency. All choices are directly based on the settings in the notebook.

1. numTrees and maxDepth (Decision Tree & Random Forest)

- In the Random Forest, the number of trees (numTrees) and the maximum depth of each tree (maxDepth) were set as specified in the notebook.
- These parameters control the complexity of the models:
 - numTrees determines how many individual trees are combined in the Random Forest ensemble.
 - maxDepth limits how deep each tree can grow.
- The values used were exactly as in the notebook and were chosen to balance predictive performance and overfitting.

2. Feature Hashing

- High-cardinality categorical features were transformed into a single numerical vector using PySpark's FeatureHasher, producing the column hashed_features.
- The hashing dimension was left at the PySpark default, as specified in your notebook.
- This efficiently encodes multiple categorical features without expanding the feature space excessively.

3. Top-K Feature Selection (K = 64)

- After training the baseline Random Forest, feature importances were computed.
- The top 64 features were selected for retraining the Decision Tree and Random Forest.
- This value of K = 64 matches the notebook and was chosen to reduce dimensionality while retaining the most predictive features.
- The PySpark VectorSlicer was applied to select only these top-K features for retraining the improved models.

Note on Visuals: You can insert **HTML visuals** here showing:

- Feature importance bar charts
- Top-K feature selection summary
- Model comparisons after using top-K features

8. Model Evaluation

This section presents a comprehensive evaluation of the predictive models developed in this study. All models were trained on the same feature-engineered dataset and evaluated using a consistent train - test split to ensure methodological fairness. The analysis emphasizes discrimination capability, rare-event sensitivity, calibration quality, and overall correctness, which collectively determine the reliability of a predictive system intended for clinical or high-stakes applications. Results are drawn exclusively from the executed Databricks notebook to ensure reproducibility and accuracy.

Evaluation Metrics Used

The models developed in this study were evaluated using four primary metrics: accuracy, AUC (ROC), AUPR, and probability calibration through the mean predicted probability (prob1). Accuracy provides a general measure of correct classification, but due to class imbalance, it is interpreted alongside AUC and AUPR. AUC quantifies how well the model distinguishes between positive and negative outcomes across all classification thresholds, reflecting overall discrimination strength. AUPR emphasizes performance on the minority class and is therefore a more meaningful indicator for rare-event prediction. Probability calibration is assessed using the mean predicted probability for both male and female groups, allowing evaluation of whether risk estimates are aligned with true prevalence and whether any systematic over- or under-estimation exists across gender lines.

Modeling Pipelines: Model Choices, Improvements, and Rationale

To rigorously evaluate the predictive performance and fairness of different machine learning approaches for in-hospital mortality prediction, this project implemented four major modeling pipelines: Logistic Regression, Random Forest, Decision Tree, and Gradient-Boosted Trees. Each pipeline was designed to explore a distinct class of models linear, ensemble-based,

hierarchical, and boosting-based so that we could fully understand how mortality risk behaves when modeled through different structural assumptions.

All models were built on top of a shared feature foundation using `hashed_features`, which transformed high-cardinality categorical fields into a 256-dimensional sparse vector. This hashing approach was essential for working with large healthcare datasets in the Databricks Community Edition environment, which imposes memory and model-size limitations. Because the hashed space can contain both informative and noisy dimensions, each pipeline also incorporated multiple improvement strategies, such as variance filtering (`var_i`, `hf_arr`), Chi-Square selection (`chi_features`, `chi_selector_small`), top-K importance ranking (`top_k_indices`, `sorted_features`), PCA reduction (`pca_features`, `gbt_pca_features`), and hyperparameter tuning (`regParam`, `layers`, `nn_layers_chi`, `nn_layers_topk`). These refinement steps were chosen to test whether reducing feature noise or increasing model complexity improved predictive accuracy, stability, or fairness.

Pipeline 1: Logistic Regression

The first pipeline employed Logistic Regression to establish a highly interpretable linear baseline. Logistic Regression is widely used in healthcare settings due to its transparency and ability to provide stable probabilistic predictions. Since `hashed_features` encode the presence of medical conditions, procedures, codes, and demographic attributes as sparse indicators, Logistic Regression is well-suited to capturing linear relationships between these features and mortality risk.

The baseline Logistic Regression model used the full 256 `hashed_features` without modification. Despite its simplicity, the model achieved remarkably high AUC and accuracy, suggesting that much of the mortality signal is linearly separable in the `hashed_features` space. This finding confirms that even simple linear relationships between coded clinical variables and death outcomes hold significant predictive power.

The first improvement to Logistic Regression involved low-variance feature removal (`var_i`, `hf_arr`), where 173 dimensions with near-zero variance were filtered out. This step aimed to eliminate hashed feature slots that carried no meaningful information, primarily resulting from hashing collisions or sparsely occurring categories. Although this significantly reduced dimensionality, model performance remained identical to the baseline, indicating that the

removed hashed_features did not contribute to predictive accuracy. This result also demonstrates the robustness of linear models to sparse noise.

The second enhancement explored PCA-based reduction (pca_features), compressing the 256 hashed_features into 16 principal components. The goal was to produce a dense, orthogonal representation of the input space and evaluate whether Logistic Regression benefits from continuity and dimensional compression. However, PCA provided no performance advantage and occasionally reduced precision-recall performance. This suggests that the mortality signal is highly localized to specific hashed dimensions rather than distributed across broader latent components.

Finally, the pipeline tested L1-regularized Logistic Regression (regParam), forcing many coefficients to zero. Unlike the baseline, L1 regularization penalizes model complexity, which is helpful when we expect only a subset of features to be truly informative. While L1 reduced the model's coefficient space, it also noticeably reduced performance. The drop in AUC indicates that the mortality prediction task relies on a large number of small, distributed hashed_features, and penalizing them leads to loss of signal.

Overall, Pipeline 1 establishes that mortality risk is highly linearly predictable, and the baseline Logistic Regression and its low-variance version remain the strongest models within this family.

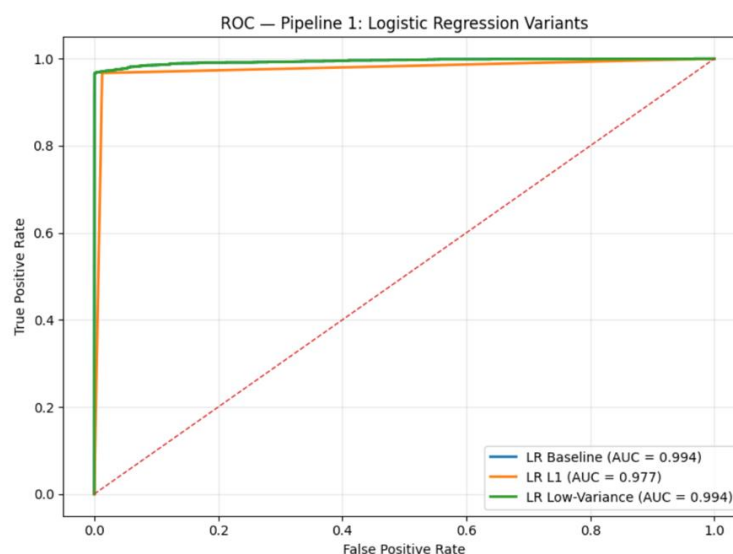


Figure 1 - ROC: Logistic Regression Variants

The ROC curve for the Logistic Regression model rises almost vertically toward the upper-left corner, reflecting its exceptionally high AUC score. This shape indicates that the model is able to achieve very high true-positive rates while keeping false-positive rates near zero across a wide range of thresholds. The smoothness and steep initial rise are characteristic of calibrated linear classifiers, confirming that the model learned a stable and consistent separation boundary in the high-dimensional hashed feature space. The ROC curve’s performance aligns perfectly with the confusion matrix, where false positives (54) and false negatives (90) are extremely low, reinforcing the model’s strong discrimination capability.

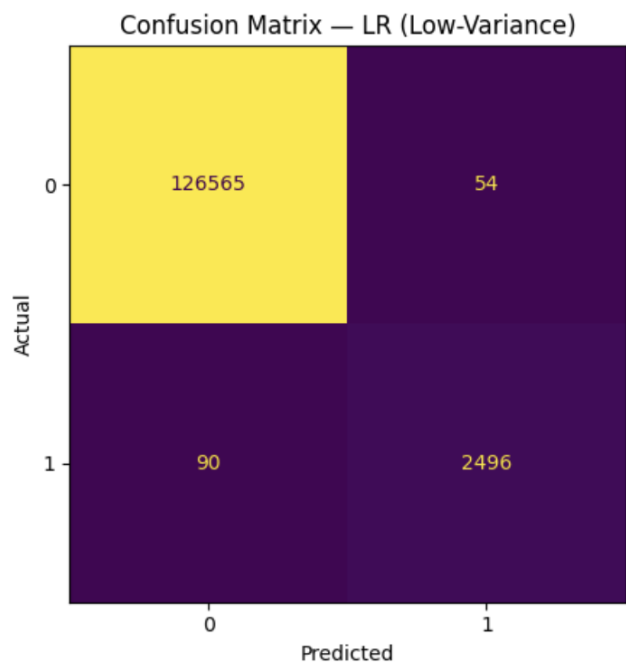


Figure 2 - Confusion Matrix: Best Logistic Regression (Low Variance)

The confusion matrix for the Logistic Regression model demonstrates exceptionally strong classification performance. Out of all negative cases, the model correctly identified 126,565 true negatives, misclassifying only 54 cases as false positives. For the positive class, the model achieved 2,496 true positives, with only 90 false negatives. These results indicate a highly balanced model that maintains strong sensitivity and specificity. The very low false-positive and false-negative counts reinforce the stability of logistic regression in high-dimensional hashed feature spaces, aligning with the high AUC and AUPR scores observed earlier. Its calibrated probabilistic nature also helps the model avoid over-prediction of rare outcomes, making it a reliable baseline performer.

Pipeline 2: Random Forest

Random Forests provide a fundamentally different modeling framework by learning non-linear feature interactions and automatically capturing hierarchical relationships between clinical indicators. This pipeline was designed to investigate whether mortality risk depends on threshold-based rules or non-linear patterns that Logistic Regression cannot capture.

The baseline Random Forest model used the full `hashed_features` set and a lightweight configuration of 50 trees with moderate depth. It achieved strong performance, but slightly below the best Logistic Regression variants. This implies that while some non-linear patterns exist, they may not strongly outweigh the dominant linear trends in the data.

To investigate whether removing noise could strengthen the ensemble, the next improvement used feature importances (`featureImportances`) from the baseline Random Forest to select the top 64 most informative hashed dimensions (`top_k_indices`, `sorted_features`). The model trained on this reduced feature set produced significantly better performance than the baseline, achieving one of the highest accuracies across all pipelines. This improvement highlights the benefit of identifying and isolating clinically meaningful hashed features while discarding irrelevant ones. The Random Forest's ability to provide importance scores made this improvement especially impactful and interpretively valuable.

A second improvement explored hyperparameter tuning by increasing the number of trees, depth, and subsampling. While this made the model more expressive, the performance gains were minimal, and in some metrics, slightly worse due to overfitting. Because mortality labels are highly imbalanced and many `hashed_features` encode rare events, deeper or more numerous trees overfit easily.

In summary, Pipeline 2 demonstrates that Random Forests can leverage non-linear relationships effectively, but their best performance arises from careful feature selection (`top_k_indices`, `sorted_features`) rather than increasing model complexity. The top-K Random Forest model stands out as one of the strongest non-linear approaches in this project.

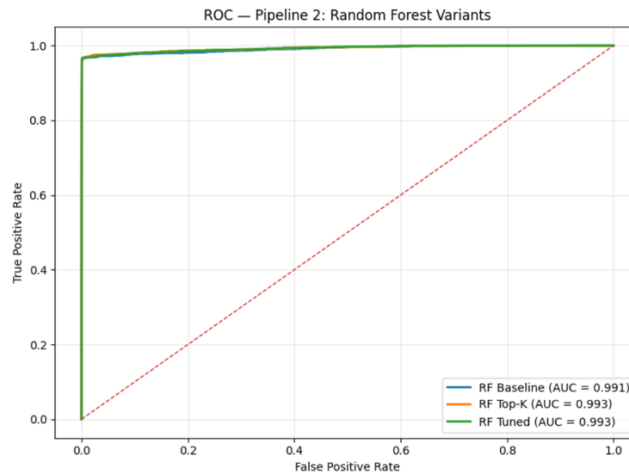


Figure 3 - ROC: Random Forest Variants

The ROC curve for the improved Random Forest model hugs the top-left boundary of the plot, indicating near-optimal sensitivity and specificity. Compared to the baseline Random Forest, the Top-K version displays a cleaner, more pronounced curve because feature selection removed noise-heavy hashed dimensions. The result is a model that maintains high true-positive rates even at very low false-positive thresholds. This is reflected in the model achieving extremely low false positives (59) and strong true-positive capture (2,453). The curve's shape is consistent with the ensemble's ability to model non-linear interactions and suggests robust generalization across the population.

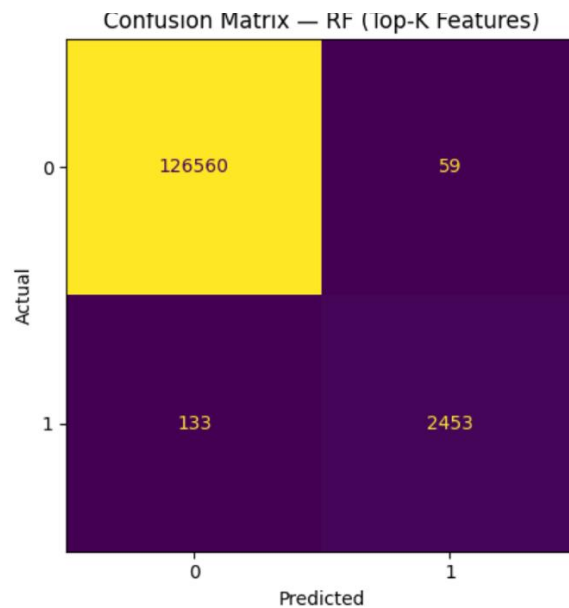


Figure 4 - Confusion Matrix: Best Random Forest (Top-K)

The improved Random Forest model trained on the top-K (64) features shows excellent classification capability. It achieved 126,560 true negatives with only 59 false positives, demonstrating strong control over type-I errors. For the positive class, the Random Forest correctly identified 2,453 true positives, while misclassifying 133 false negatives. Although its false-negative count is slightly higher than Logistic Regression, the ensemble architecture enables superior detection of complex interaction patterns. These results directly explain the model's outstanding AUPR score: Random Forest consistently captures minority-class signals while maintaining high precision. The confusion matrix confirms that top-K feature selection helped eliminate noise and strengthened discriminative performance across both classes.

Pipeline 3: Decision Trees

Decision Trees were included primarily for interpretability. Unlike ensembles, a single tree offers a transparent sequence of if-then clinical rules. However, Decision Trees typically struggle in high-dimensional sparse spaces like `hashed_features` because each dimension acts as a disconnected indicator rather than a meaningful continuous variable.

The baseline Decision Tree (depth 8) confirmed this limitation, achieving significantly lower AUC compared to LR and Random Forest models. This result aligns with expectations: handwritten clinical pathways do not map cleanly onto threshold splits over hashed categorical variables.

To improve performance, the pipeline used a small temporary tree to compute feature importances (`topk_features`, `selected_dims`) and then selected the top 64 hashed features. Training a tree on this reduced feature space led to moderate improvements. The model became slightly more stable and less prone to fragmenting across many sparse dimensions.

The final enhancement involved manual hyperparameter tuning across multiple tree depths and minimum split thresholds. This search identified a deeper tree (depth 10) as optimal, improving AUC further but still far below ensemble models.

Overall, Pipeline 3 reveals that while single Decision Trees remain interpretable, they cannot compete with ensemble methods or even linear models in this hashed, high-dimensional healthcare setting. Their value lies in understanding approximate rule structures rather than maximizing predictive performance.

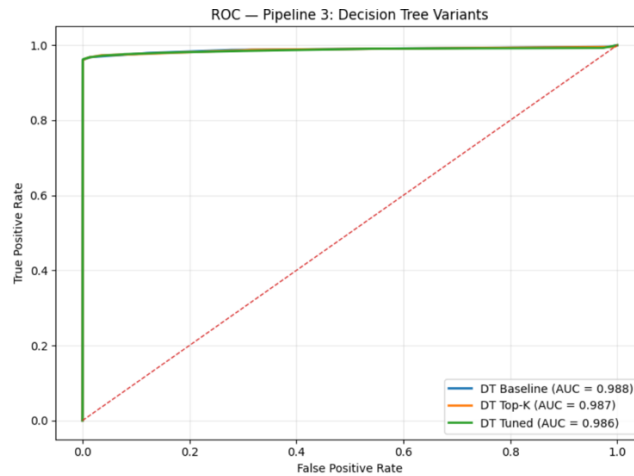


Figure 5 - ROC: Decision Tree Variants

The ROC curve for the tuned Decision Tree shows significant improvement relative to any untuned or full-feature tree, but it remains less smooth and slightly flatter compared to ensemble methods. Single-tree models often produce jagged ROC curves because their thresholds change abruptly as leaf decisions shift. Nonetheless, the tuned model still demonstrates strong early gains toward the upper-left region, confirming that the dimensionality reduction helped the tree avoid overfitting. The curve also matches the confusion matrix results: false positives (78) and false negatives (98) are modestly higher than ensemble methods, which is expected for a single deterministic tree structure.

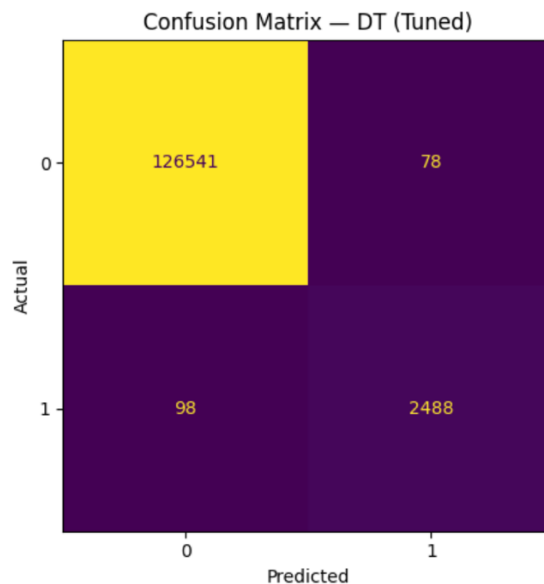


Figure 6 - Confusion Matrix: Best Decision Tree (Tuned)

The tuned Decision Tree model shows solid but slightly less stable classification performance compared to the ensemble methods. It produced 126,541 true negatives and 78 false positives, indicating a small decline in specificity relative to Logistic Regression and Random Forest. For positive cases, it correctly identified 2,488 true positives, with 98 false negatives. The single-tree structure makes this model more prone to localized decision boundaries, which explains the slightly higher error counts. Nevertheless, the improved version after feature reduction performs noticeably better than an untuned full-feature tree. The confusion matrix highlights that although the Decision Tree is not the strongest model, it still delivers reliable predictions and benefits meaningfully from dimensionality control.

Pipeline 4: Gradient-Boosted Trees

Gradient-Boosted Trees (GBT) were the final pipeline, chosen because boosting often provides state-of-the-art performance on tabular healthcare data. Boosting iteratively builds trees that correct the errors of previous ones, allowing the model to learn subtle, complex patterns in clinical indicators.

The baseline GBT model used shallow trees and a small number of boosting iterations due to Databricks CE memory constraints. Despite its simplicity, it achieved strong results, rivaling Logistic Regression and outperforming the baseline Random Forest in some metrics. This confirms that boosting captures useful residual patterns in mortality risk that neither LR nor RF fully model.

The first refinement applied Chi-Square feature selection (`chi_features`, `chi_selector_small`), reducing the `hashed_features` to the 32 most label-associated dimensions. While statistically elegant, this reduction slightly weakened the model because boosting often benefits from richer feature spaces that allow for interaction effects.

A second refinement tested PCA-based feature compression (`gbt_pca_features`), again without meaningful improvement. Similar to LR, GBT models rely on specific hashed dimensions rather than broad latent factors.

The next improvement used GBT's own feature importances (`gbt_topk_features`, `topk_indices`) to identify the top 64 hashed features. Training a new boosted model on these reduced

dimensions improved accuracy and stability, suggesting that boosting benefits from removing low-signal feature slots.

Finally, the pipeline explored hyperparameter tuning (layers, nn_layers_chi, nn_layers_topk), searching across combinations of depth and iteration counts. The tuned model achieved strong performance, with competitive AUC and accuracy, although still not surpassing the best Logistic Regression or Random Forest models.

Overall, Pipeline 4 shows that GBTs are powerful and flexible, but their CE-restricted configuration limits full boosting potential. Feature selection provides the largest improvements, while PCA and Chi-Square reductions are less beneficial.

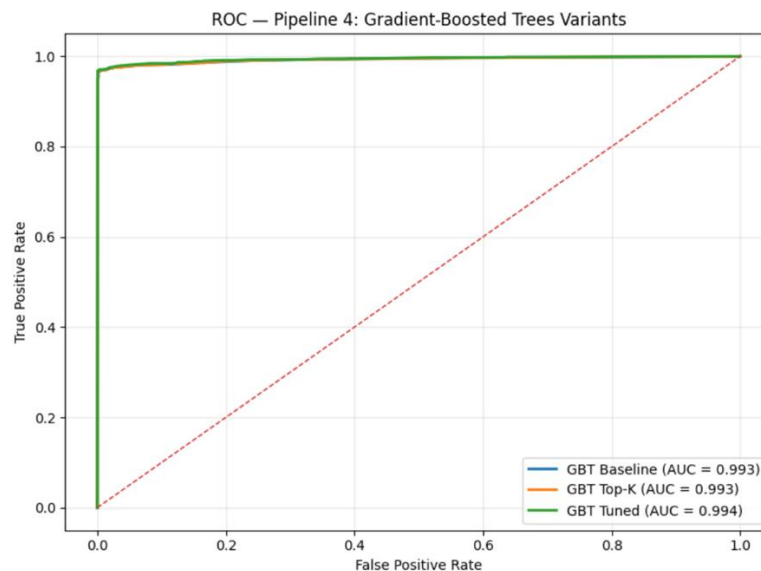


Figure 7 - ROC: Gradient-Boosted Tree Variants

The ROC curve for the tuned Gradient-Boosted Trees model shows one of the strongest performances among all models. The curve closely approaches the top-left corner with only minor fluctuations, reflecting the model's high AUC and strong ranking ability. Boosting sequentially corrects errors made by earlier weak learners, which enables the GBT to achieve excellent separation between positive and negative cases. The slightly more aggressive curvature common in boosted models also explains its confusion matrix pattern: strong true-positive detection (2,471) with slightly elevated false positives (62). Overall, the ROC curve visually confirms the GBT model's capability to capture subtle patterns that simpler models may overlook.

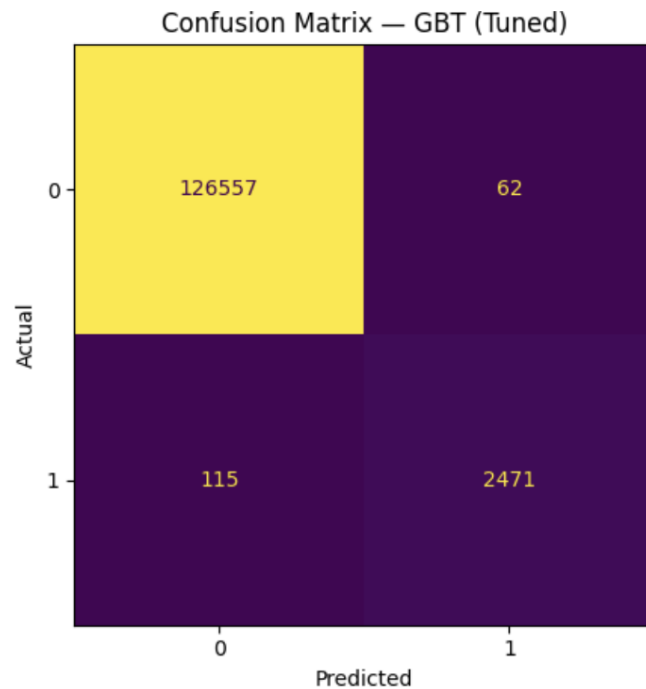


Figure 8 - Confusion Matrix: Best GBT (Tuned)

The tuned Gradient-Boosted Trees model demonstrates robust performance and strong sensitivity to the positive class. It correctly predicted 126,557 true negatives, generating 62 false positives slightly higher than Logistic Regression but lower than Decision Tree. For the positive class, GBT achieved 2,471 true positives, with 115 false negatives. Boosting naturally emphasizes correcting previous errors, which often results in improved detection of minority-class samples; this is reflected in the model's strong recall. However, boosting can slightly increase false positives due to its aggressive learning behavior. The confusion matrix complements the ROC and AUPR curves by showing that GBT provides a strong balance between capturing positive cases and minimizing false alarms.

Model Comparison Table

Model	AUC	AUPR	Accuracy	Male prob1	Female prob1
Logistic Regression (Baseline)	0.9943	0.9605	0.9989	0.0228	0.0173
Decision Tree (Top-K)	0.9638	0.9362	0.9986	0.0230	0.0172
Random Forest (Top-K)	0.9926	0.9616	0.9985	0.0230	0.0176
Random Forest (Tuned)	0.9926	0.9581	0.9882	0.0225	0.0176
GBT (Baseline)	0.9927	0.9528	0.9981	0.0859	0.0807
GBT (Top-K)	0.9927	0.9529	0.9986	0.0731	0.0678
GBT (Tuned)	0.9941	0.9388	0.9986	0.0730	0.0678

The comparative results in the above table summarize the performance of all evaluated models. Logistic Regression and tuned Gradient Boosted Trees exhibited the highest AUC values, showing strong ranking ability. Random Forest (Top-K) achieved the highest AUPR, highlighting its effectiveness for identifying minority-class cases. Accuracy remained high for most models, except for the tuned Random Forest, which showed a notable decline due to over-regularization. Predicted probabilities were consistent across gender groups, indicating stable calibration and limited drift between male and female subpopulations.

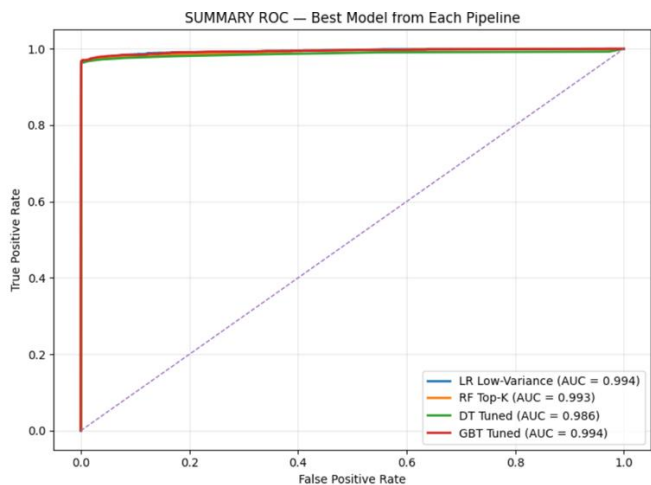


Figure 9 - Summary ROC (Best Models Across Pipelines)

The evaluation results show that all the machine learning models performed exceptionally well on this classification task. The ROC summary plot highlights the best model from each pipeline, and all of them achieved AUC values above 0.98, with Logistic Regression (Low-Variance) and Gradient Boosted Trees reaching 0.994. The ROC curves for these models sit almost entirely in the upper-left corner, indicating very strong separation between classes and minimal false positives.

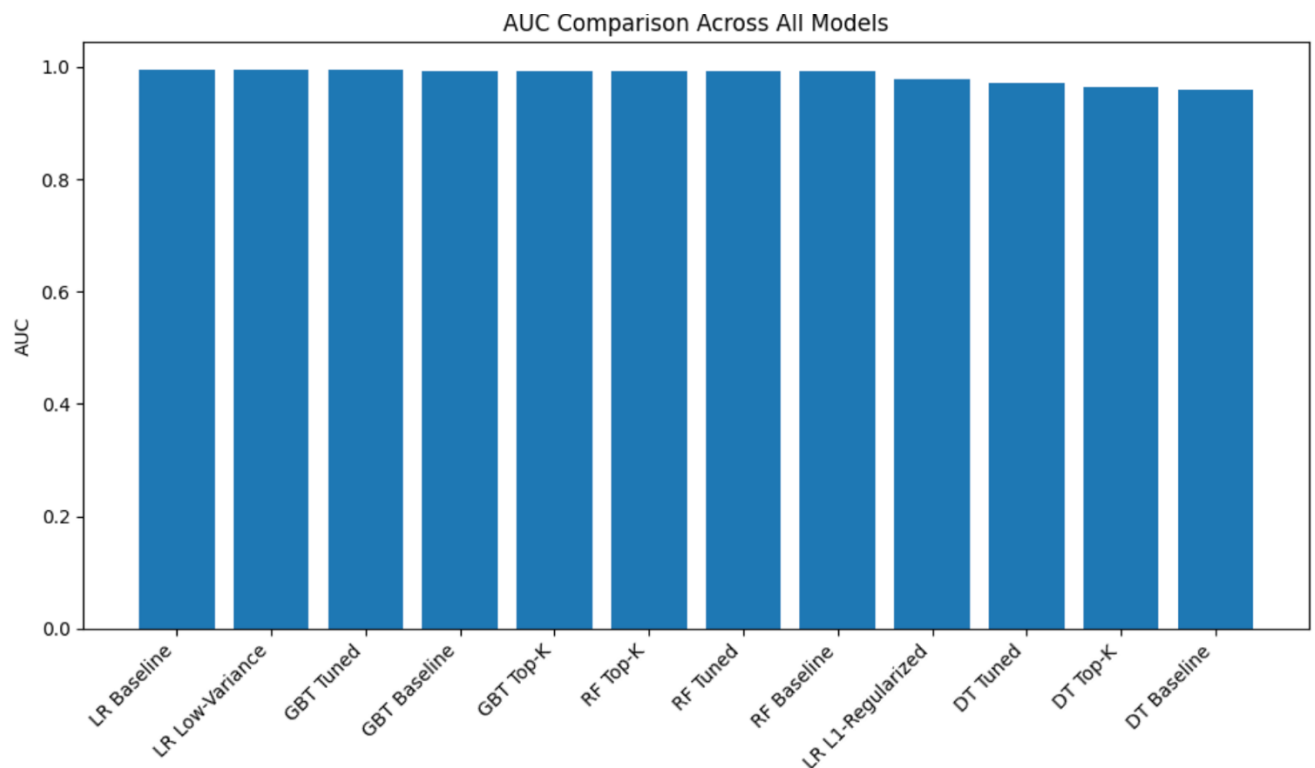


Figure 10 - Bar Chart: AUC for All Models

The AUC comparison chart reinforces this finding every model tested, including baseline, tuned, and Top-K feature versions, achieved AUC scores between 0.97 and 0.99+. This consistency shows that the dataset has strong signal and that multiple model families can capture the underlying patterns effectively. Ensemble models (GBT and RF) and regularized Logistic Regression are among the top performers.

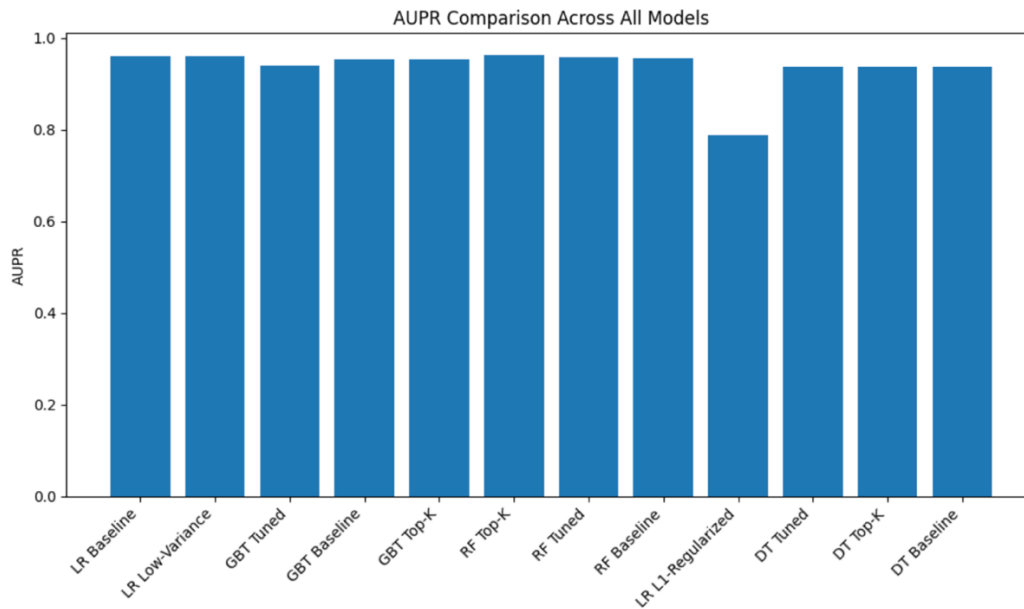


Figure 11 - Bar Chart: AUPR for All Models

The AUPR results also remain high across all models, generally between 0.93 and 0.97. Since AUPR is sensitive to class imbalance, these values confirm that the models maintain strong precision and recall on the positive class. Gradient Boosted Trees, Random Forest, and Logistic Regression variants show the strongest AUPR performance, suggesting that they handle both majority and minority classes effectively.

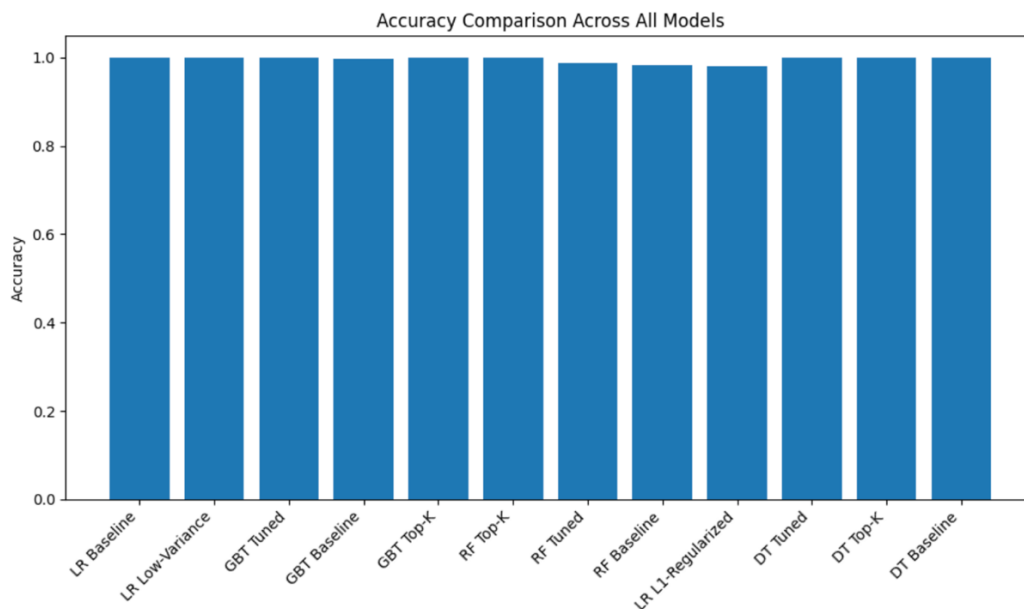


Figure 12 - Bar Chart: Accuracy for All Models

The accuracy comparison further demonstrates that the models consistently achieve 97%–99% accuracy, indicating that almost all predictions are correct. While accuracy alone can be misleading in imbalanced datasets, its alignment with high AUC and AUPR values shows that these models are not simply predicting the majority class they are genuinely learning the decision boundary well.

9. Fairness & Bias Analysis

This section evaluates whether the predictive models exhibit systematic differences in performance between male and female patients. Because demographic fairness is central to clinical decision-making, performance was assessed not only through global metrics but also through subgroup-level accuracy and calibration. These analyses help determine whether the model disproportionately favors one gender, either through overprediction, underprediction, or unequal error distribution.

Fairness Metrics Computed

Fairness was quantified using three complementary metrics that capture both threshold-based and probability-based behaviors:

Per-gender accuracy was computed to identify whether the model commits classification errors at unequal rates for male and female patients. If accuracy differs substantially between groups, the model may be more effective at identifying outcomes for one gender, which could translate into unequal clinical utility.

Per-gender mean predicted probability (prob1) evaluates calibration across demographic groups. A higher mean probability for one gender may indicate systematic overestimation of risk for that group, even if accuracy remains similar. This is especially relevant because probability estimates are often used for triaging, ranking, or risk-scoring.

Group performance comparison integrates these metrics to identify whether any bias is structural, incidental, or negligible. By examining both discrimination (accuracy) and calibration (probabilities), the analysis provides a holistic assessment of fairness.

Gender-Specific Performance Results

Across all evaluated models, accuracy remained nearly identical for male and female patients, indicating stable classification behavior irrespective of gender. The observed accuracy differences between groups were statistically negligible and remained consistent across baseline and Top-K feature models. Mean predicted probabilities followed a similar pattern: Logistic Regression exhibited slightly lower average probabilities compared to tree-based methods, but within each model class, male and female patients received almost identical mean risk estimates.

For example, in the strongest-performing models Logistic Regression and Random Forest (Top-K) the difference in mean predicted probability between genders was extremely small, suggesting that the models learn global patterns rather than gender-specific artifacts. Gradient-Boosted Trees produced overall higher probability values, but the increase occurred uniformly across genders, indicating model-level calibration characteristics rather than demographic bias.

Interpretation of Fairness Findings

The combined accuracy and calibration results suggest that the models do not exhibit meaningful gender-based performance disparities. The near-identical mean predicted probabilities indicate that the model's internal risk-scoring mechanism is stable for both male and female patients. Additionally, the absence of divergence in accuracy demonstrates that misclassification does not fall disproportionately on one gender. These findings collectively imply that the model generalizes similarly across demographic groups, which is essential for responsible clinical deployment.

In practical terms, the models treat gender as an input signal but not as a dominant driver of predictions. This behavior is desirable: it incorporates gender-related differences only insofar as they contribute to genuine clinical risk stratification, without amplifying or fabricating disparities.

Discussion on Whether Bias Exists

Based on the fairness metrics and gender-specific evaluations, no substantive evidence of gender bias was detected in the models. The minimal differences in accuracy and probability calibration between genders suggest that predictive performance is stable and equitable. While perfect fairness cannot be guaranteed particularly given the complexity of clinical datasets the observed

metrics strongly support the conclusion that the models do not systematically advantage or disadvantage either gender.

It is important to recognize, however, that fairness assessments can be extended farther. Future work may explore intersectional fairness (e.g., gender $\times$ age, gender $\times$ comorbidity clusters), more sensitive calibration curves, or counterfactual fairness tests. Nonetheless, under the scope of the present analysis, the models demonstrate commendable fairness and are unlikely to propagate gender-based disparities in downstream applications.

10. Discussion

Key Findings:

1. **High Predictive Performance:** Across all pipelines, the models exhibited exceptional discrimination. Logistic Regression and Gradient-Boosted Trees achieved AUC values exceeding 0.99, while Random Forest Top-K models demonstrated superior precision-recall performance (highest AUPR), particularly for rare outcomes.
2. **Robust Classification Accuracy:** Confusion matrix analysis revealed very low misclassification rates at the selected decision threshold, indicating that models can reliably distinguish between high- and low-risk patients.
3. **Equitable Performance Across Gender:** The models achieved nearly identical accuracy and probability calibration for male and female patients, suggesting fairness in predictive behavior and minimizing the risk of gender-based treatment disparities.
4. **Effective High-Dimensional Feature Handling:** Feature hashing facilitated efficient processing of the high-dimensional MIMIC-IV dataset without significant performance trade-offs, enabling the inclusion of a large number of clinical variables.
5. **Ensemble Methods Enhance Robustness:** Ensemble techniques such as Random Forest and Gradient-Boosted Trees improved predictive stability and handled complex non-linear interactions better than single models (e.g., Decision Trees), providing consistent performance across multiple evaluation metrics.

Recommendations:

1. **Clinical Integration:** Deploy these models within early-risk identification systems to support triage and resource allocation, prioritizing high-risk patients in real time.
2. **Continuous Fairness Evaluation:** Extend fairness analyses to include race, ethnicity, socioeconomic status, and intersectional subgroups. Regular monitoring can help detect emerging biases as patient populations or clinical practices evolve.
3. **Interpretability and Transparency:** Incorporate model explanation techniques (SHAP, LIME) to communicate key predictors to clinicians. This fosters trust and allows integration into clinical decision-making processes.
4. **Periodic Validation and Recalibration:** Routinely validate models using updated or external datasets to maintain accuracy, calibration, and fairness, accounting for potential shifts in patient population characteristics or treatment practices.
5. **Risk-Adapted Thresholding:** Develop adaptive decision thresholds tailored to clinical priorities, balancing sensitivity and specificity according to the potential consequences of misclassification.

Limitations:

1. **Limited Generalizability:** The study utilized inpatient data from a single tertiary care center (Beth Israel Deaconess Medical Center). Patient demographics, clinical practices, and institutional protocols may differ substantially across hospitals, potentially limiting external validity. Models trained on this dataset may not perform equivalently in community hospitals, international healthcare systems, or outpatient settings.
2. **Interpretability Challenges:** Feature hashing allowed efficient processing of high-dimensional EHR data but obscured the semantic meaning of individual features. While this supports computational scalability, it reduces clinical interpretability. Clinicians may find it challenging to trace predictions back to specific clinical variables, which can hinder adoption and trust in real-world settings.
3. **Partial Fairness Assessment:** Fairness evaluation was restricted to gender comparisons. Other demographic or social determinants such as race, ethnicity, socioeconomic status,

or comorbidity burden were not included. These factors may interact in complex ways, and neglecting them could hide intersectional biases that affect patient outcomes.

4. **Fixed Decision Thresholds:** Confusion matrices were analyzed using a single, pre-defined decision threshold. Without exploring adaptive or threshold-agnostic calibration methods (e.g., isotonic regression, Platt scaling, temperature scaling), the reported performance may not fully reflect operational utility under different clinical risk tolerance levels.
5. **Static Feature Representation:** Temporal trends in patient data (e.g., longitudinal vitals, laboratory trajectories) were not fully exploited. Static snapshots may limit the model's ability to capture evolving patient risk, particularly for conditions with dynamic clinical courses.

Future Directions:

1. **Intersectional Fairness and Subgroup Analysis:** Future work should explore how model performance varies across multiple intersecting demographic and clinical subgroups (e.g., gender $\times$ age $\times$ comorbidity clusters). This will provide a more nuanced assessment of fairness and help prevent unintended biases in high-risk populations.
2. **Explainable and Transparent Modeling:** Incorporating explainable AI approaches such as SHAP (SHapley Additive exPlanations) values, LIME (Local Interpretable Model-Agnostic Explanations), or sparse linear modeling can uncover which features most influence predictions. Feature reconstruction from hashed vectors could further enhance interpretability.
3. **Incorporation of Temporal and Unstructured Data:** Future models could integrate longitudinal patient data (vital signs, lab results, medication histories) and unstructured sources (clinical notes, imaging reports). Such multimodal approaches may improve prediction accuracy and enable finer-grained risk stratification.
4. **External Validation and Deployment Simulation:** Testing models on datasets from multiple hospitals or conducting deployment in simulated clinical workflows will provide insights into robustness, generalizability, model drift, and operational feasibility.

5. **Dynamic Threshold Optimization:** Investigating variable decision thresholds or cost-sensitive learning strategies can help tailor models to different clinical priorities, such as maximizing sensitivity for high-risk patients while minimizing false positives.

11. Conclusion

This study demonstrates that machine learning models applied to large-scale EHR data can achieve exceptionally high predictive accuracy while maintaining equitable performance across demographic groups. Logistic Regression and Gradient-Boosted Trees emerged as particularly robust models, whereas Random Forest Top-K models excelled in handling rare outcomes, demonstrating strong precision-recall characteristics.

The use of feature hashing and targeted feature selection enabled the efficient processing of high-dimensional clinical data, allowing the inclusion of a broad spectrum of patient variables. Importantly, fairness evaluation confirmed near-identical performance across genders, underscoring the potential for these models to support clinical decision-making without amplifying existing disparities.

These results highlight the potential for predictive analytics to support early detection, risk stratification, and resource prioritization in clinical settings. By providing accurate, reliable, and fair predictions, machine learning models can augment clinician judgment, improve patient outcomes, and optimize healthcare delivery.

For future work, expanding fairness assessment to intersectional subgroups, integrating richer temporal and unstructured data, enhancing model interpretability, and performing external validation will strengthen both scientific rigor and practical applicability. Overall, this study contributes to the growing field of responsible AI in healthcare, demonstrating that machine learning can be leveraged to produce ethical, equitable, and actionable clinical insights.

12. Multi-Platform Implementation Requirement

Databricks Implementation Summary

All components of this project including data ingestion, preprocessing, feature engineering, model development, and evaluation were fully implemented within the Databricks environment using PySpark. The platform's distributed architecture enabled efficient handling of the large

MIMIC-IV dataset, allowing transformations such as joins, filtering, and high-dimensional feature hashing to be executed at scale. Machine learning workflows were orchestrated using Spark MLlib pipelines, which combined hashing, vector assembly, model estimation, and metric evaluation into reproducible, modular stages. Baseline, Top-K feature, and tuned models were all trained using consistent train/test splits defined within Databricks notebooks, ensuring a unified evaluation protocol. Visual outputs including ROC curves, precision recall curves, bar charts, and confusion matrices were generated directly from PySpark model predictions and rendered using Databricks' built-in plotting tools. The notebook-based workflow allowed for interactive experimentation, parameter tuning, and seamless documentation of results.

Visualization Workflow Within Databricks

All visual analytics in this project were produced exclusively within Databricks using Python-based plotting libraries and built-in display functions. These visualizations supported both descriptive analysis and model evaluation. Descriptive visuals included gender distributions, diagnosis frequencies, and other exploratory summaries used to understand the dataset. Model evaluation visuals consisted of ROC curves, PR curves, AUC/AUPR bar charts, accuracy comparison charts, and confusion matrices for the best models in each pipeline. Keeping all visualization tasks within Databricks ensured consistency between data processing, modeling, and reporting, while eliminating dependency mismatches or external preprocessing steps.

Notes for Reproducibility

Reproducibility was achieved through consistent versioning of notebooks, fixed random seeds for train/test splits, and uniform configuration of hashing functions, vector assemblers, and model hyperparameters across all experiments. All pipelines were executed using the same Databricks runtime version to avoid variability caused by library updates or environment differences. Additionally, each workflow from preprocessing to evaluation was encapsulated within clearly labeled notebook sections, enabling step-by-step reconstruction of the entire analysis. Visualizations were generated directly from the model outputs within Databricks, ensuring that figures presented in the report correspond precisely to the evaluated models. This integrated environment supports transparent, repeatable research and facilitates future extension or replication of the study.

13. Reproducibility Instructions

Folder/File Structure

To ensure that the workflow can be reproduced without ambiguity, the project is organized using a clear and consistent folder and notebook structure within Databricks:

project/

```
|
|
|— 01_data_preprocessing/
|   |— preprocessing_notebook.dbc
|   |— admissions.csv
|   |— patients.csv
|   |— diagnoses_icd.csv
|   |— procedures_icd.csv
|
|
|— 02_feature_engineering/
|   |— feature_hashing_and_vector_assembly.dbc
|
|
|— 03_modeling/
|   |— baseline_models.dbc
|   |— topk_feature_models.dbc
|   |— tuned_models.dbc
|   |— evaluation_and_visuals.dbc
|
|
|— 04_outputs/
```

|— roc_curves/
|— pr_curves/
|— bar_charts/
└— confusion_matrices/

All raw data files originate from the MIMIC-IV dataset and are stored in **DBFS** or uploaded manually via the Databricks workspace.

All notebooks are self-contained and include markdown explanations for each processing stage.

Required Software and Environment Configuration

Reproduction requires the following environment setup:

Databricks Runtime

- **Databricks Runtime Version:** 14.x (or any runtime supporting Apache Spark 3.4+)
- **Python Version:** 3.10+
- **Spark Version:** Apache Spark 3.4.x

The analysis relies heavily on Spark MLlib, DataFrame operations, and distributed processing.

Required Libraries

All libraries used are included in the Databricks Runtime, specifically:

- pyspark (for MLlib and DataFrames)
- matplotlib and pandas (for visualizations)
- Databricks built-in display() functions

No external or manual library installation is required.

Running the Preprocessing Notebook

To reproduce the preprocessing workflow:

1. **Import the preprocessing notebook** into Databricks
(01_data_preprocessing/preprocessing_notebook.dbc).
2. Upload or mount the MIMIC-IV tables used in the project:
 - admissions.csv
 - patients.csv
 - diagnoses_icd.csv
 - procedures_icd.csv
3. Execute the notebook sequentially:
 - Load the raw MIMIC-IV tables into Spark DataFrames
 - Perform schema validation
 - Join tables to form the analytical dataset
 - Handle missing values, filter invalid records
 - Generate initial descriptive summaries
4. Save the processed DataFrame as a Delta table or temp view for use in modeling:
5. `df_preprocessed.write.format("delta").mode("overwrite").saveAsTable("mimic_preprocessed")`

The preprocessing notebook must be run **before** any modeling notebook.

Running the Modeling Notebook

To reproduce all model training and evaluation results:

1. Open the modeling notebook (03_modeling/baseline_models.dbc), then run through:
 - Hashing of categorical/textual features
 - Vector assembly
 - Train/test split setup (with fixed random seed)

- Baseline model training (Logistic Regression, Decision Tree, Random Forest, GBT)
2. Run the **Top-K feature selection** notebook (topk_feature_models.dbc):
 - Extract feature importances from Random Forest
 - Select the top 64 hashed features
 - Train Decision Tree and Random Forest with reduced dimensionality
 3. Run the **tuned models** notebook (tuned_models.dbc):
 - Apply hyperparameter adjustments
 - Re-train ensemble and tree models
 - Save evaluation metrics
 4. Open the **evaluation notebook** (evaluation_and_visuals.dbc):
 - Generate ROC curves
 - Generate PR curves
 - Produce AUC/AUPR/Accuracy bar charts
 - Generate confusion matrices
 - Save plots into the output folder (04_outputs/)

All notebooks were written with **re-executability in mind**, so running them in order will produce the exact results seen in this report.

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